

Matei Mihai Alexandru

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Portfolio

PROFILE

Third-year CTAI student at Howest and full-time AI developer at an automation startup. I like building AI that works with real hardware, from ESP32 firmware to cloud backends.

EXPERIENCE

AI Developer

Oblivion Labs · West Flanders, Belgium · July 2026 – Present

- PDF plan parser that sorts thousands of architectural drawings and extracts floor areas (Python).
- Automated social media posting across platforms with a lightweight LLM, cutting time per batch from 30 to 4 minutes.
- Quoting assistant that turns spec sheets into catalogue proposals using an LLM (Next.js, Azure).

Creative Technology Lead

Kurkumama Group (KKM) · Kortrijk, Belgium · March 2026 – May 2026

- Led AI tool adoption across four venues: reporting dashboards and a staff-training chatbot.

Freelance Web Developer

Self-employed · Websites for local businesses · 2025 – Present

PROJECTS

Drone-Based Airfield Lighting Inspection

Industry project for Intersoft Electronics · Team of 4 · 2026

- Detects and maps runway lights from night drone footage (YOLO + SAHI) with 96–97% identification accuracy.
- Built the Azure backend (FastAPI, Container Apps, Bicep) and the frontend.

360° Immersive Room & ESP32 Scent Diffusers

Howest University · Team project, product owner · 2025 – 2026 · github.com/xChocapic/experiential-learning

- Voice input becomes an 8K 360° panorama with matching sound and scent in under a minute (.NET, Azure).
- Designed and programmed four ESP32 scent diffusers in C, controlled over MQTT, with 3D-printed housings.

Smart Lamp System

Howest University · Solo project · 2025 · github.com/howest-mct/2024-2025-projectone-ctai-mihaialexmatei

- A Raspberry Pi 5 runs a custom YOLOv8 gesture model and controls two ESP32 lamps over MQTT.

EDUCATION

Bachelor of Applied Sciences – Creative Technology and AI (CTAI)

Howest University of Applied Sciences · Kortrijk, Belgium · 2024 – 2027 (expected)

- Coursework: robotics and edge AI, computer vision, generative AI, MLOps, embedded systems.

Technical Informatics – High School Diploma

Colegiul Național Gheorghe Șincai · Bucharest, Romania · 2020 – 2024

- Graduated 1st of class with a 4-year GPA of 9.7/10, at one of the top 5 high schools in Romania.

SKILLS & INTERESTS

Programming	Python, C / C++, C# / .NET, SQL
Hardware	ESP32 (ESP-IDF, Arduino), Raspberry Pi, soldering, 3D printing
AI & ML	YOLO training, PyTorch, OpenCV, generative AI pipelines, RAG
Platforms & Tools	Azure (Functions, IoT Hub, OpenAI), Docker, FastAPI, MQTT, Linux
Interests	Robotics, embedded systems, homelabbing, self-hosted infrastructure
Languages (spoken)	Romanian (native), English (fluent)